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Large Language Models for Radiology QA: Benchmarking and Error Taxonomy on RadQA

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***Abstract*—Radiology reports are a critical source of clinical information but are often lengthy, variable in structure, and difficult for non-radiologists to navigate. Question answering (QA) systems can help clinicians rapidly retrieve relevant findings, yet existing evaluations of extractive QA on radiology reports rely heavily on aggregate metrics such as exact match (EM) and F1, which obscure important failure modes. In this work, we benchmark modern open large language models (LLMs), such as LLaMA 3.1-8B, LLaMA 3.2-3B, and Mistral-Small-22B, on the RadQA dataset under zero-shot, few-shot, and fine-tuned settings. We report performance using EM, F1, and a sentence-level measure (SM), and introduce a structured error taxonomy that disaggregates incorrect predictions into eight categories, including hallucination, superset, subset, and abstention errors. Our results show that fine-tuning substantially outperforms in-context prompting, with LLaMA 3.1-8B achieving 61% EM, 77% F1, and 85% SM, surpassing previously reported baselines on RadQA. The taxonomy reveals that fine-tuning not only raises accuracy but also eliminates hallucinations and reduces false positive and false negative abstentions, concentrating residual errors in boundary-level categories such as superset and subset. These findings demonstrate the value of combining standard metrics with taxonomy-driven analysis to better understand model behavior and highlight future directions, including reasoning-oriented LLMs and hybrid extractive–generative approaches, for building reliable clinical QA systems.**

***Index Terms*—large language model, question answering, radiology, EHR, error taxonomy**

I. Introduction

Radiology studies inform decisions that affect diagnosis, treatment planning, and patient outcomes. Each year, millions of imaging studies are conducted, and their findings are documented in narrative radiology reports. These reports serve as the primary means of communication between radiologists and other clinicians. However, their length, variable structure, and use of highly specialized terminology often make it difficult for non-radiologist physicians to quickly locate the information most relevant to their patients. As a result, critical insights may be overlooked [1]–[4] or take longer to retrieve [1], [2], slowing clinical workflows [1]–[3] and increasing cognitive burden on healthcare providers [1], [2]. Recent advances in natural language processing (NLP) have created new opportunities to support clinicians in navigating unstructured electronic health records (EHRs), including radiology reports [5]–[10]. Among these methods, question answering has emerged as a particularly intuitive approach: it allows a clinician to pose a targeted question and receive a precise answer. Such a system could help physicians quickly confirm or rule out findings, improving efficiency and potentially enhancing patient care. To facilitate research in this area, Soni et al. [11] introduced the RadQA dataset, a collection of 3,074 question–answer pairs (6,148 including multiple spans for the same answer) grounded in radiology reports. Uniquely, RadQA's questions were crafted from the clinical referral indications of the exams – i.e. the actual information needs of physicians who ordered the imaging. This design ensured that the questions target real-world clinical queries (e.g. "Is there evidence of pulmonary embolism?"), and it organically produced unanswerable questions in cases where the report did not address the query. Each question is answered with a specific span from the radiologist's Findings or Impression section, providing a concise yet complete answer phrase (often more than a single entity) that can span multiple lines of text. By framing radiology report understanding as an extractive QA task, RadQA directly links to clinical decision-making: a robust QA system could help clinicians rapidly retrieve key findings (or determine that the report does not contain an answer), thereby improving workflow and patient care.

Extractive QA in the radiology domain poses considerable challenges and holds significant clinical importance. Radiology reports are a form of unstructured EHR data known for their diverse, semantically complex content. They contain domain-specific language (e.g. medical abbreviations, anatomical terms) and often require expert interpretation. Questions about a report frequently demand integration of information across sentences and the application of medical knowledge or reasoning beyond the text itself. In an analysis of RadQA, for example, a large majority of the questions were found to require substantial medical background knowledge to answer. This makes the task challenging even for advanced NLP models. Moreover, the inclusion of unanswerable questions introduces a critical real-world scenario: a system must recognize when a report does not contain the requested information. Handling such cases correctly is as important as finding answers, since producing an unwarranted answer can mislead clinicians. Notably, RadQA was the first clinical QA dataset to explicitly include unanswerable questions – a feature inspired

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by open-domain QA benchmarks that stressed the importance of "knowing what you don't know". A QA model must therefore decide when to abstain from answering in addition to extracting correct answer spans when they do exist. These characteristics (specialized language, cross-sentence reasoning, and the presence of questions with no answer) underline both the difficulty and the clinical relevance of the extractive QA task in radiology.

Despite the clear need and the introduction of RadQA as a benchmark, there remain significant gaps in the literature. Initial evaluations on RadQA reported only moderate success: the best baseline model (a transformer fine-tuned on MIMIC-III radiology reports) achieved an F1 score of 63.6%, while expert human performance was over 90.0% [11]. This large performance gap highlights the difficulty of the task and the inadequacy of early methods. Subsequent research has improved performance by leveraging more powerful and domain-tuned models. For instance, Lehman et al. [12] conduct a comprehensive study of foundation models for radiology QA and show that a fine-tuned BioClinicalRoBERTa model can reach roughly 75.0% F1 on RadQA, representing the state-of-the-art for this dataset. This result reinforces a trend: domain-specific encoder models (e.g. RoBERTa-based) tend to outperform generic large language models (LLMs) on extractive clinical QA tasks [13]. However, even with 75% F1, the machine performance still lags notably behind human-level comprehension, and the RadQA challenge is far from solved. Moreover, many of these efforts have focused on traditional fine-tuning of encoder models, without exploring the capabilities of the latest generative LLMs in this domain. There is a lack of systematic benchmarking of how modern large language models – such as the recent LLaMA and Mistral models – perform on radiology QA, especially under zero-shot or few-shot prompting conditions. It remains an open question whether advanced LLMs can close the gap on a task that demands both domain knowledge and precise extraction.

Another under-addressed aspect of prior work is the handling of unanswerable questions and the analysis of model failure modes. Most existing studies report only aggregate metrics like EM and F1 to summarize QA performance. While these metrics are standard and useful, they do not tell the full story of how models fail. Indeed, recent evaluations have noted that token overlap metrics like F1, though reasonable for span-based QA, may not be suitable for all extractive QA scenarios and answer types [14]. A model's answer can be wrong in different ways: in some cases, the model produces an answer where the true answer was "no answer", or, conversely, fails to produce any answer when one was present. It might include too much irrelevant text beyond the gold answer (token superset error), omit part of the correct answer (token subset error), or overlap only partially with the gold span (token overlap error). Without a detailed analysis, these failure modes are conflated into the same drop in EM/F1. Furthermore, the challenge of unanswerable questions is often glossed over – many models will attempt to answer every question, a tendency that was explicitly addressed in open-domain QA by introducing unanswerable questions in datasets like SQuAD 2.0 [15], [16], but in the clinical domain this behavior has not been thoroughly analyzed. To build reliable radiology QA systems, we need to understand not just how often models are wrong, but how they are wrong. In particular, identifying whether errors come from inability to locate relevant facts, misunderstanding the question, or not handling the "no-answer" case can guide more targeted improvements.

In this paper, we aim to advance the state of radiology report QA by evaluating a range of modern LLMs on the RadQA dataset and by performing a fine-grained error analysis of their performance. Specifically, our contributions are as follows:

1) **Comprehensive Evaluation of Diverse LLMs**: We conduct an extensive evaluation of several state-of-the-art large language models – including LLaMA 3.1-8B, LLaMA 3.2-3B, and Mistral-small-22B on RadQA. These models span different sizes and architectures, allowing us to assess the impact of model scale and design. We evaluate each model under both zero-shot and few-shot prompting conditions, as well as in a fine-tuned setting where the model is specialized on the RadQA training set. This comprehensive experiment is, to our knowledge, the first to benchmark the newest generation of open LLMs on radiology QA, and it enables direct comparisons between in-context learning and fine-tuning approaches.
2) **Standard Metric Results and Unanswerable Question Handling**: We report model performance using standard QA metrics (EM and token-level F1) on both answerable and unanswerable questions. This includes a rigorous comparison to prior published results on RadQA. In particular, we examine how well these models handle questions with no answer. By measuring performance on unanswerable cases separately (e.g. the frequency of correct "no answer" predictions), we shed light on the models' ability (or inability) to abstain when appropriate. This analysis addresses a gap in the literature, as previous studies have rarely isolated performance on unanswerable questions or explored strategies to improve it.
3) **Structured Error Analysis with a Detailed Taxonomy**: We introduce a novel error taxonomy to categorize the types of mistakes models make on this extractive QA task. Each non-Exact-Match prediction is assigned exactly one of seven error types; Gold No-Answer (model predicts an answer when it should not), Prediction No-Answer (the ground-truth has an answer span, but the model gave no answer), Hallucination (prediction is not in the clinical note.), No Overlap (the prediction has no overlap with the gold answer at all), Token Superset (the prediction strictly contains the gold answer text plus additional tokens), Token Subset (the prediction is contained inside the gold answer, but incomplete), or Token Overlap (the prediction intersects with the gold answer but neither is a subset of the other). By

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systematically classifying errors in this manner, we gain fine-grained insights beyond the aggregate F1 score. For example, we can determine if a model commonly adds irrelevant information to an otherwise correct answer (superset errors) or if it frequently misses relevant details (subset errors), etc. This kind of analysis helps reveal predominant failure modes and highlights specific areas for improvement (such as better distinguishing relevant from irrelevant text, or improving recall of all critical findings in an answer). We also include a sentence match accuracy (how often the predicted answer is at least in the same sentence as the gold answer). This allows us to isolate near-misses (superset, subset, overlap, and no overlap but in the same sentence) from potentially more egregious incorrect answers. To our knowledge, this is the first work to present such an organized error analysis for radiology report QA. It provides a deeper understanding of why models fail, complementing the traditional evaluation of how often they fail.

II. Related work

The task of clinical QA has received increasing attention in biomedical NLP due to its potential to improve access to information in unstructured EHRs. Among various subdomains, radiology report comprehension has emerged as a critical focus area because of the high density of clinically relevant information and the complexity of interpreting narrative radiology findings. While several QA datasets have been developed in the clinical domain, such as emrQA [17] and PubMedQA [18], the RadQA dataset stands out for its emphasis on real-world, patient-specific questions derived from radiology report referral indications and its explicit inclusion of unanswerable questions.

A. BERT-based Clinical QA Models on RadQA:

Early work on RadQA by Soni et al. [11] benchmarked a range of transformer-based models fine-tuned on clinical and biomedical corpora, including ClinicalBERT [19] and BioBERT [20]. These models achieved moderate performance, with Bio+Discharge Summary BERT reaching an F1 score of 63.6%, highlighting the difficulty of the extractive QA task in this domain. ClinicalBERT, though pretrained on discharge summaries, struggled to generalize well to radiology language, which is often more concise and fragmented.

Subsequent work by Lehman et al. [12] explored the use of larger and more specialized encoder-based models, including BioClinicalRoBERTa and ClinicalLongformer, showing that domain-specific pretraining combined with extended input contexts can significantly improve extractive QA performance. In particular, a Longformer model fine-tuned on RadQA achieved a new state-of-the-art F1 score of 75.0%, demonstrating the value of extended context modeling and domain adaptation. Despite these gains, these models still rely heavily on large-scale supervision and are limited by their fixed context size and architecture constraints.

B. Large Language Models for Clinical Question Answering:

The advent of large language models (LLMs) such as GPT-3 [21], GPT-4 [22], PaLM [23], and LLaMA [24] has transformed open-domain question answering. These models, with billions of parameters and broad pretraining, demonstrate strong zero- and few-shot reasoning capabilities across tasks. However, their applicability to domain-specific extractive QA—particularly in radiology—remains underexplored. In many cases, LLMs exhibit hallucinations or overly general responses when applied to structured medical data, especially when the answer must be tied precisely to a span within a source document.

Lehman et al. [12] conducted a comprehensive study evaluating several foundation models—including GPT-3, GPT-4, and ChatGPT—on clinical QA tasks. Their findings show that while general-purpose LLMs perform surprisingly well in low-data settings, smaller, fine-tuned clinical models can still outperform them on extractive QA benchmarks like RadQA. For instance, they report that fine-tuned domain models consistently deliver more precise span predictions and handle answerability with greater reliability, even as GPT-based models struggle with unanswerable questions or overgenerate responses.

While models like Med-PaLM [25] have pushed the boundaries of generative QA in medicine, there is a notable lack of systematic evaluation of the latest open models (such as LLaMA 3.1/3.2, and Mistral-small) on extractive clinical datasets like RadQA. Moreover, most prior studies have not examined how these LLMs perform under different prompting strategies (zero-shot vs. few-shot) or compared their effectiveness to fine-tuned variants. This gap motivates our present study.

C. Evaluation and Error Analysis in Extractive QA:

Evaluation in clinical QA has primarily relied on traditional metrics such as EM and token-level F1 [15]. While useful, these metrics can obscure the nuanced ways in which a model may succeed or fail. For example, a model might provide a partial but clinically relevant answer and still be penalized as incorrect. Few works have gone beyond aggregate metrics to conduct detailed error analysis in extractive QA tasks.

In the context of RadQA, annotators have previously used qualitative labels such as "missed relevant info" or "insufficient specificity" to categorize inter-annotator disagreements [11]. However, no study to date has applied a structured error taxonomy to model predictions. This leaves open the question of how LLMs fail on extractive clinical QA, especially when faced with unanswerable questions or ambiguous spans. Categorizing these errors (such as identifying when a predicted span is a token superset, token subset, contains no overlap with the reference, or is even in the right "neighborhood" of the correct answer) can yield deeper insights into model behavior and reliability.

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III. METHODS

A. Dataset

RadQA [11] consists of 1,009 radiology reports paired with 6,148 clinically motivated questions and span-based gold answers. Each question is derived from the clinical referral indication associated with the imaging study and is answered, when possible, by a contiguous span from the report text. Importantly, a portion of the questions (1,754, 28.5%) are explicitly labeled as unanswerable, meaning the radiology report contains no information relevant to the physician's query. This feature reflects real-world clinical practice, where a clinician's information need is not always addressed in the radiologist's interpretation.

In addition to overall dataset size, we further characterize RadQA in terms of answerability and answer span length, as these properties directly influence task difficulty and model behavior. For answerable questions, gold answer spans vary substantially in length, ranging from short atomic phrases to multi-clause clinical descriptions. The average answer span length is 16.2 tokens, with a median of 7 tokens, indicating a skewed distribution with a long tail of longer spans. By definition, unanswerable questions have an answer length of zero. This variability highlights the dual challenge of correctly determining answerability while also extracting precise boundaries for answers of heterogeneous length.

For all experiments, we followed the official RadQA split: the training set was used for fine-tuning, and the test set was held out for evaluation. The same test set was used consistently across the zero-shot, few-shot (5-, 10-, 15-shot), and fine-tuned model evaluations to ensure comparability of results.

In the few-shot prompting setting, we manually curated a set of 15 representative examples from the training split (Table I) to serve as in-context demonstrations. Selection was guided by coverage across answerability (both answerable and unanswerable cases), span length (short atomic spans and longer clause-level answers), and clinical content (device/line placement, interval change, negative findings, anatomy variety).

To enable controlled comparisons across different few-shot sizes, we used a nested selection protocol: the 5-shot subset is contained within the 10-shot subset, which is contained within the 15-shot subset. This approach allows us to isolate the incremental effect of adding demonstrations, while keeping the composition of smaller sets consistent.

We did not apply additional preprocessing or filtering to the dataset. Each report was presented to the models in its entirety, including Findings, Impression, and any additional sections. This design preserved all potentially relevant context for answering the question and mirrored the natural input length encountered in clinical practice.

B. Task Formulation

We frame the problem as an extractive QA task over radiology reports. Given the full text of a clinical note R and a clinician-posed question Q, the goal is to predict an answer span A that is a contiguous sequence of tokens from R that fully answers Q. If no such span exists in the report, the correct output is an empty string, indicating that the question is unanswerable based on the report's content.

$$A \subseteq R \quad \vee \quad A = \varnothing. \tag{1}$$

During inference, we instructed the LLMs to return exactly the string "None" when they determined that no answer was present. For evaluation purposes, these "None" predictions were mapped to empty strings so they matched the gold-standard representation of unanswerable questions in the RadQA dataset.

Our evaluation explicitly considers both answerable and unanswerable cases. When both the gold answer and the model prediction are empty, this is treated as an Exact Match category, since the model correctly abstained from answering. In contrast, misclassified unanswerable cases fall into two error categories in our taxonomy (Section III-D):

- **Gold No Answer (GNA):** the gold answer is empty but the model predicts a span.
- **Pred No Answer (PNA):** the gold answer contains a span but the model outputs no answer.

This design ensures that models are assessed not only on their ability to extract correct spans when an answer exists but also on their ability to abstain appropriately when an answer is absent.

C. Model Setup

We evaluated three LLMs: LLaMA 3.1-8B, LLaMA 3.2-3B, and Mistral-Small-22B. All evaluated models are general-purpose open LLMs rather than radiology- or clinically specialized language models. This choice allows us to assess how far broadly trained, reproducible models can be adapted to extractive radiology QA through prompting and task-specific fine-tuning, independent of domain-specific pretraining. We used the base variants rather than instruction-tuned (chat) versions, in order to directly assess their raw extractive capabilities. Each model was evaluated under three paradigms:

- **Zero-shot prompting:** the model is given only the task description and the test instance without any examples.
- **Few-shot prompting:** the model was given $k = 5, 10, 15$ representative examples from the RadQA training set.
- **Fine-tuning:** the models were trained on the RadQA training set using LoRA applied to both attention and feed-forward layers.

For inference, we used the Ollama framework [26], which provided a standardized interface for model loading and execution. This allowed us to run zero-shot and few-shot prompting experiments reproducibly across all model families.

For fine-tuning, we employed the Unsloth [27] and Hugging Face Transformers [28] libraries. Hugging Face tokenizers were used to ensure consistent text preprocessing, while Unsloth enabled efficient parameter-efficient fine-tuning (LoRA) with 4-bit quantization. This hybrid setup ensured both resource efficiency and comparability across models.

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TABLE I

TEN FEW-SHOT EXAMPLES, CHOSEN FROM RADQA TRAINING SET AND USED IN THE PROMPTING SETUP.

User Message	Assistant Message
Clinical Note: REASON FOR EXAMINATION: Evaluation of patient with severe acute mitral regurgitation, intra-aortic balloon pump, interval change and line placement evaluation. Portable AP chest radiograph was compared to prior study obtained at 5:56 p.m. on [**2154-3-19**]. The Swan-Ganz catheter inserted through the femoral approach terminates at the level of right main pulmonary artery just below the aortic arch, unchanged in position. No intra-aortic balloon pump demonstrated. The pacemaker leads terminate in right atrium and right ventricle. The mild cardiomegaly predominantly with enlargement of the left ventricle is unchanged. There is mild volume overload but no overt pulmonary edema. Small bilateral pleural effusions are most likely present and there is no evidence of pneumothorax. **Question:** Where do we see the swan line positioned?	Terminates at the level of right main pulmonary artery just below the aortic arch
Clinical Note: DATE OF SERVICE: [**2124-6-10**] at 17:25 hours. HISTORY: Constipation x3 days with elevated white blood cell count. COMPARISON: None. FINDINGS: There is persistent blunting of the right costophrenic angle with a significant non-dependent meniscus which may indicate a loculated effusion or chronic scarring. No free air is detected in the abdomen. Scattered air-fluid levels are noted. Overall, there is a gentle paucity of bowel gas. The air-fluid levels noted are most likely colonic. Despite the given history, no significant fecal loading is identified with the exception at the rectal vault. No organomegaly or ascites detected. There is significant multilevel degenerative change throughout the lumbar spine with a dextroconcave curvature present. **Question:** Can we see any excess air in the abdominal cavity?	No free air is detected in the abdomen
Clinical Note: REASON FOR EXAMINATION: Follow-up of the patient intubated due to sepsis. Portable AP chest radiograph was compared to two studies obtained on [**12-18**], [**2129**]. The ET tube tip is approximately 5 cm above the carina. The NG tube tip has been advanced in the interim giving the absence of the sidehole above the diaphragm. The cardiomediastinal silhouette is stable. Extensive opacities within the lungs are present. Interval development of vascular engorgement is seen that might be consistent with volume overload/mild pulmonary edema. Note is made that the left costophrenic angle was excluded from the field of view. **Question:** How is the progression of pneumonia on the chest x ray?	None
Clinical Note: HISTORY: Followup pleural effusion. FINDINGS: In comparison with study of [**6-11**], there is probably some increase in the initial amount of right pleural effusion seen layering out along the right lateral chest wall on the prior study. Mild opacification at the left base is consistent with a small effusion. Bibasilar atelectasis is seen. Cardiac silhouette remains enlarged. Coarse interstitial markings could reflect chronic lung disease, elevated pulmonary venous pressure, or both. **Question:** Did the plueral effusion reduce in size?	Some increase in the initial amount of right pleural effusion
Clinical Note: HISTORY: Trauma. Assess for fracture. AP AND LATERAL THORACIC SPINE: Portable technique. Upper thorax is obscured on the lateral views. The vertebral body heights and disc spaces appear preserved without evidence of fracture or malalignment. Please see chest x- ray reports for other intrathoracic findings. AP AND LATERAL LUMBAR SPINE: Portable technique. The vertebral body height and disc spaces appear preserved without evidence of fracture or malalignment. The visualized pelvis appears within normal limits. **Question:** Are there any fractures of the lumbar spine?	Without evidence of fracture
Clinical Note: IMPRESSION: Unchanged left basal atelectasis; however, concomitant consolidation cannot be ruled out. Persistently elevated right hemidiaphragm. No new acute cardiopulmonary process. **Question:** Was there any change over the interval in left facial tumor?	None
Clinical Note: PFI REPORT PFI: Normal bowel gas pattern. No evidence of free air. **Question:** How is the area under the diaphragm?	Normal bowel gas pattern. No evidence of free air
Clinical Note: HISTORY: Trauma. Assess for fracture. AP AND LATERAL THORACIC SPINE: Portable technique. Upper thorax is obscured on the lateral views. The vertebral body heights and disc spaces appear preserved without evidence of fracture or malalignment. Please see chest x- ray reports for other intrathoracic findings. AP AND LATERAL LUMBAR SPINE: Portable technique. The vertebral body height and disc spaces appear preserved without evidence of fracture or malalignment. The visualized pelvis appears within normal limits. **Question:** How are the thoracic vertebrae?	Vertebral body heights and disc spaces appear preserved without evidence of fracture or malalignment
Clinical Note: PREOP CHEST ON [**12-13**] HISTORY: Pre-op AVR. Evaluate for effusion. **Question:** How is the pericardium area?	None
Clinical Note: IMPRESSION: Unchanged left basal atelectasis; however, concomitant consolidation cannot be ruled out. Persistently elevated right hemidiaphragm. No new acute cardiopulmonary process. **Question:** Is the chest x ray showing any interim abnormalities?	Unchanged left basal atelectasis; however, concomitant consolidation cannot be ruled out. Persistently elevated right hemidiaphragm

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TABLE II
SYSTEM, USER, AND ASSISTANT MESSAGES IN A k-SHOT PROMPT.

System message	"You are a highly knowledgeable medical doctor and an expert Natural Language Processing (NLP) system. You are provided with a Clinical Note and a Question. Your task is to extract only the exact portion of the Clinical Note that contains the answer to the Question. If the answer is not present in the Clinical Note, respond with 'None'. Do not explain your answer. Do not include any additional text—just output the span or 'None'."
User message (Example 1)	**Clinical Note:** [note of Ex 1] **Question:** [question of Ex 1]
Assistant message (Example 1)	[gold answer of Ex 1]
...	
User message (Example k)	**Clinical Note:** [note of Ex k] **Question:** [question of Ex k]
Assistant message (Example k)	[gold answer of Ex k]
User message (Test Example)	**Clinical Note:** [note from test set] **Question:** [question from test set]

Experiments were conducted on a server equipped with dual AMD EPYC 7352 CPUs (96 threads total) and 8 NVIDIA A100-SXM4 GPUs (80 GB each), running CUDA 13.0.

All experiments used deterministic generation settings (temperature = 0.0). For fine-tuning, each model configuration was trained using a single run with a fixed random seed (seed = 3407), and no averaging across multiple runs was performed. The maximum generation length was set to accommodate the longest possible span in the dataset. Model outputs were post-processed by trimming leading/trailing whitespace and replacing "None" with an empty string to match the gold annotation format.

Prompting Strategy: In the zero- and few-shot settings, a fixed system instruction specified the task, followed by either several example QA pairs (few-shot) or no examples (zero-shot).

In the few-shot condition, the system prompt was followed by k = 5, 10, 15 manually selected examples from the RadQA training set. These examples were chosen to be representative and comprehensive, covering a range of question types, answer lengths, and answerability conditions. As Table II shows, each example consisted of:

- User message: containing the full clinical note (or the relevant Findings/Impression sections) and the question.
- Assistant message: containing only the gold-standard answer span, or "None" for unanswerable cases.

Ten examples are shown in Table I. The final user message in the prompt sequence contained the clinical note and question of the evaluation instance, whose answer was to be predicted.

In the zero-shot condition, only the system prompt was provided, immediately followed by the evaluation instance's clinical note and question, without any in-context examples.

D. Evaluation

Model outputs were normalized and evaluated using the official SQuAD v2.0 evaluation protocol, which extends the original SQuAD metric formulation [15] to explicitly support unanswerable questions [16]. We used the publicly released evaluation script (`evaluate-v2.0.py`) [1] provided by the SQuAD authors, which applies standard text normalization including lowercasing, punctuation removal, article removal, and whitespace normalization. EM and token-level F1 were computed following this protocol, with correct abstentions (both gold and prediction empty) treated as exact matches. The normalization steps applied were:

1) Lowercasing all text.
2) Removing punctuation characters.
3) Removing the English articles "a", "an", and "the".
4) Collapsing multiple spaces into a single space.

Unanswerable questions are represented in the gold data as an empty string. To ensure consistency, any model output of "None" was replaced with "" prior to evaluation. This aligns with the gold standard for unanswerable cases, ensuring that correct abstentions are rewarded.

Evaluation Metrics: We evaluated models using three complementary metrics:

- **EM:** The percentage of predictions that exactly match a gold answer based on SQuAD matching policy.
- **Token-level F1:** The harmonic mean of precision and recall computed over tokens in the normalized prediction and gold answer. Both EM and F1 are computed per question and micro-averaged over the dataset.
- **SM:** A sentence-level measure of alignment between gold and predicted spans. Each clinical note was segmented into sentences and each gold/pred span mapped to the set of sentence indices it covered. SM is defined as:
  1) If both gold and prediction are empty:
  $$SM = 100\% \tag{2}$$
  2) Else:
  $$SM = \frac{|G \cap P|}{max(|G|, |P|)} \times 100, \tag{3}$$

where G is the gold sentence set and P the predicted sentence set. One could imagine a QA system that returns the sentences containing the answer instead of the answer span itself. SM then measures the accuracy of such (relaxed) approach. It also helps to identify if the LLM is "close" to the right answer within the document.

To account for minor lexical variation, we introduced a **token-difference threshold** when assigning predictions to sentences. If the token difference between a predicted span and a strict subset of a sentence was equal or less than threshold, the sentence index was included in P. The threshold was further adjusted for very short predictions: if $\frac{|\text{pred tokens}|}{4} < 1$, the threshold was reduced stepwise until it reached zero. This ensured that small lexical differences (e.g., "situation" vs. "sutuation") did not prevent correct sentence alignment.

Together, SQuAD EM and F1 scores capture exact and token-level accuracy, while SM provides a complementary

[1]The evaluation script is publicly available at: https://github.com/rajpurkar/SQuAD-explorer/blob/master/evaluate-v2.0.py

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measure of whether the model located the correct sentence(s), even if the extracted boundaries were imperfect.

For questions with multiple gold answers, the score for a prediction is computed against each gold span, and the maximum value is retained. This max-over-golds rule is applied consistently across all three metrics (EM, F1, and SM), ensuring that a prediction is credited with its best possible match when multiple gold-standard spans are available.

In addition to overall scores, we further categorized each prediction into one of eight answer types to provide a fine-grained breakdown of model performance on answerable versus unanswerable questions.

Error Taxonomy: To obtain fine-grained insights into model behavior beyond EM and F1, we designed a structured error taxonomy. Each prediction is deterministically assigned to exactly one of eight mutually exclusive categories.

Because each question in RadQA may have two gold answers (one each in the Findings and Impressions sections), we compared the prediction against all gold spans and selected the gold yielding the higher F1 score.

Formally, let G denote the selected gold span, P the predicted span, and R the source report. Set operations $(\cap, \subset, \supset)$ are defined over tokenized spans. The taxonomy is defined as follows:

1) **Exact Match:** Prediction exactly matches the gold span (including the case where both are empty).
$$P = G, \tag{4}$$
2) **Gold No Answer (GNA):** Gold is empty but the model outputs a non-empty span.
$$(G = \varnothing) \wedge (P \neq \varnothing), \tag{5}$$
3) **Pred No Answer (PNA):** Gold span is non-empty but the model predicts nothing.
$$(G \neq \varnothing) \wedge (P = \varnothing), \tag{6}$$
4) **Hallucination:** Predicted span is not present in the source report (after thresholded comparison).
$$P \not\subseteq R, \tag{7}$$
5) **No Overlap:** Prediction and gold share no tokens.
$$(G \neq \varnothing) \wedge (P \neq \varnothing) \wedge (P \cap G = \varnothing), \tag{8}$$
6) **Token Superset:** Prediction strictly contains the gold span plus additional tokens.
$$(G \neq \varnothing) \wedge (P \neq \varnothing) \wedge (G \subset P), \tag{9}$$
7) **Token Subset:** Prediction is strictly contained within the gold span.
$$(G \neq \varnothing) \wedge (P \neq \varnothing) \wedge (P \subset G), \tag{10}$$
8) **Token Overlap:** Prediction and gold overlap partially, but neither fully contains the other.
$$(P \cap G \neq \varnothing) \wedge (P \not\subset G) \wedge (G \not\subset P). \tag{11}$$

This taxonomy ensures that every prediction falls into exactly one category, capturing both **boundary errors** (superset, subset, overlap), **answerability errors** (GNA, PNA), and **severe failures** (no overlap, hallucination), while rewarding correct abstentions as **Exact Match**.

While EM and token-level F1 are standard for extractive QA, they do not fully capture semantic or clinical correctness. A prediction may be clinically acceptable despite minor boundary deviations, or clinically misleading despite token overlap. We therefore complement these metrics with SM and taxonomy-based error analysis to better reflect practical reliability.

IV. Results

Table III summarizes aggregate performance for the evaluated models, with the best-performing scores per metric highlighted in **bold** for ease of comparison. Across all settings, fine-tuning provides the largest improvements. For LLaMA 3.1-8B, performance increases from 25.7% EM, 55.4% F1, and 63.4% SM in the zero-shot setting to 33.9% EM, 63.6% F1, and 71.9% SM with 15-shot prompting, and further to almost 61% EM, 77% F1, and 85% SM after fine-tuning. LLaMA 3.2-3B follows the same trend, with fine-tuned results of 55.7% EM, 74.3% F1, and 82.2% SM. These outcomes confirm that task-specific supervision yields substantially stronger extractive QA performance than in-context prompting alone. These results, while still below human performance on RadQA (estimated above 90.0% F1), substantially narrow the gap compared to prior automated systems. In particular, the fine-tuned LLaMA 3.1-8B model achieves 77.0% F1, surpassing the previously reported best performance of approximately 75.0% F1 from BioClinicalRoBERTa on RadQA. Few-shot prompting improves over zero-shot, but the effect is modest and insufficient to close the gap with fine-tuned models.

Mistral-Small-22B behaved differently and underperformed relative to both LLaMA models. In zero- and few-shot conditions, EM and F1 scores are very low (e.g., 4.9% EM and 13.0% F1 in zero-shot), though SM is higher (around 33–40%), indicating that the model often identifies the correct region of the report but fails to extract accurate spans. After fine-tuning, SM rises substantially to 75.3%, but EM and F1 remain low relative to the LLaMA models. This suggests that Mistral benefits from supervision at the sentence level yet still struggles with precise boundary extraction. In summary, Mistral consistently underperforms relative to the LLaMA models, with weak EM and F1 scores and a persistent tendency to abstain, even after fine-tuning.

To better understand model behavior, we applied the structured error taxonomy described in Section III-D. Table IV provides a more detailed view of model behavior through the error taxonomy. In this table, favorable outcomes (higher Exact Match and lower hallucination and abstention errors) are highlighted in **bold** to facilitate interpretation. Fine-tuning shifts the error distribution strongly toward the left side of the taxonomy spectrum, with higher proportions of Exact Match, Superset, and Subset categories, and sharply reduces severe errors. Hallucinations, which are particularly problematic in clinical contexts because they introduce content not present in the report, are eliminated entirely after fine-tuning in both LLaMA models. Similarly, both GNA and PNA errors decrease, reflecting improved calibration. Before

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TABLE III
EVALUATION METRICS ACROSS MODELS AND SETTINGS.

LLM Model	Setting	EM(%)	F1(%)	SM(%)
LLaMA 3.1-8B	0-shot	25.73	55.35	63.35
	5-shot	27.36	58.38	67.58
	10-shot	34.20	63.31	70.76
	15-shot	33.88	63.62	71.86
	fine-tuned	**61.07**	**76.99**	**85.01**
LLaMA 3.2-3B	0-shot	20.03	39.07	44.16
	5-shot	18.89	43.40	50.85
	10-shot	23.78	47.91	54.83
	15-shot	22.80	48.32	56.35
	fine-tuned	55.70	74.32	82.16
Mistral-Small-22B	0-shot	4.89	13.00	40.00
	5-shot	3.75	7.78	33.51
	10-shot	4.56	8.63	34.23
	15-shot	4.72	10.29	36.23
	fine-tuned	11.00	35.31	75.29

fine-tuning, Mistral is dominated by PNA errors (over 60%), showing a strong tendency to abstain even when answers exist. Fine-tuning alleviates this issue, although performance still lags behind the LLaMA family. Overall, fine-tuned LLaMA models deliver the strongest results, hallucination errors are eliminated, and handling of unanswerable questions improves markedly compared to zero- and few-shot prompting.

It is important to distinguish between EM in Table III and the Exact Match category in Table IV. EM is calculated using the standard SQuAD v2.0 script (without a token-difference threshold beyond SQuAD normalization), whereas taxonomy-based Exact Match is assigned after applying the same normalization *plus* a small token-difference threshold. As a result, taxonomy Exact Match typically tends to be slightly higher than EM, or very close to it.

A strong relationship also emerges between SM and the sum of Exact Match, Superset, Subset, and Overlap categories in Table IV, since these represent predictions that at least identify the correct sentence even if boundaries are imperfect. This highlights SM as a complementary relaxed measure of correctness that captures sentence-level localization beyond token-level scoring.

Taken together, these results show that fine-tuned LLaMA models achieve state-of-the-art performance on RadQA, with the 8B variant performing best. Although the gap with human performance remains substantial, particularly in boundary selection, the improvements in EM, F1, and SM highlight the importance of domain adaptation. In contrast, Mistral-Small underperformed across all settings, with fine-tuning failing to produce competitive extractive accuracy despite some gains in sentence-level matching. The error taxonomy further revealed that boundary misalignment remains the predominant failure mode, while unanswerable question handling and hallucination control improve markedly with fine-tuning.

V. DISCUSSION

The results highlight several key observations. Fine-tuning consistently outperforms zero-shot and few-shot prompting across all models. The largest gains are seen for LLaMA 3.1-8B, which achieves state-of-the-art performance on RadQA with 77% F1, indicating that task-specific supervision remains crucial for extractive radiology QA. Few-shot prompting improves performance compared to zero-shot, but its effect is limited and insufficient to close the gap with fine-tuned models. These findings reinforce the importance of domain adaptation when applying large language models to clinical QA.

The error taxonomy in Table IV provides additional insights into how models fail and how their errors evolve with supervision. Fine-tuning shifts the distribution of errors toward more favorable categories such as Exact Match, Superset, and Subset, while reducing severe errors including Hallucination, GNA, and PNA. Hallucinations are particularly concerning in clinical settings because they introduce content not present in the source note; their elimination after fine-tuning is an encouraging sign of improved reliability. Among GNA and PNA errors, the relative risk is context-dependent: GNA represents a false positive where the model invents an answer, which can mislead clinicians, while PNA represents a false negative where the model abstains despite an available answer, which reduces utility but is less dangerous. Importantly, these two error types have different clinical implications. GNA represents a false positive, where the model produces an answer despite the report not containing relevant information; such errors pose higher clinical risk because they may mislead downstream decision-making. In contrast, PNA represents a false negative, where the model abstains despite an available answer; while undesirable, this behavior is generally less harmful clinically, as it reflects uncertainty rather than fabrication [29]. Our results show that fine-tuned LLaMA models reduce both error types substantially, while hallucinations—the most severe failure mode—are entirely eliminated after fine-tuning. These findings underscore the importance of jointly evaluating answerability and extraction accuracy in clinical QA. The LLaMA models show balanced reductions in both categories, whereas Mistral initially exhibits very high PNA, indicating a strong tendency to abstain. Although fine-tuning mitigates this behavior, its performance remains lower than the LLaMA family. Our error taxonomy further highlights distinct patterns of model behavior. Superset errors indicate that models often err on the side of verbosity, producing answers that include the gold span but extend beyond it. Subset errors, by contrast, reflect under-extraction, where models identify only part of the correct span. These complementary tendencies show that models are frequently close to the correct answer but struggle with precise boundary alignment. Overlap errors capture partial correctness when the prediction intersects the gold span without fully containing it, while No Overlap represents a more severe mislocalization. Hallucination remains the most dangerous category clinically, as it introduces content not present in the source report. Hallucinated answers were observed primarily in zero-shot and few-shot settings, where models attempted to respond to clinically plausible questions despite the absence of explicit supporting evidence in the report. In some cases the hallucination was relatively harmless, reflecting information in the report but with terminological or word order differences. However, this is still not ideal: answers should be grounded in the report

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TABLE IV
ERROR TAXONOMY DISTRIBUTION (%).

LLM Model	Setting	Exact_Match	T_Superset	T_Subset	T_Overlap	No_Overlap	GNA	PNA	Hallucination
LLaMA 3.1-8B	0-shot	25.90	32.08	5.05	3.91	5.21	11.89	11.89	4.07
	5-shot	27.52	33.22	7.00	3.75	5.37	13.36	6.19	3.58
	10-shot	34.04	28.83	8.96	3.42	4.56	7.82	9.28	3.09
	15-shot	33.71	30.62	7.17	4.40	4.72	9.45	6.84	3.09
	fine-tuned	**60.59**	13.84	9.45	2.12	5.05	6.51	**2.44**	**0.00**
LLaMA 3.2-3B	0-shot	20.03	17.92	6.19	3.75	7.00	12.87	29.15	3.09
	5-shot	18.73	22.64	12.05	1.79	6.51	15.31	12.54	10.42
	10-shot	23.78	21.50	11.24	3.09	7.33	14.50	9.77	8.79
	15-shot	22.80	25.41	8.96	3.09	5.86	13.19	11.89	8.79
	fine-tuned	**55.37**	17.43	8.63	2.28	4.07	7.33	**4.89**	**0.00**
Mistral-Small-22B	0-shot	28.50	5.54	4.72	2.77	1.47	1.14	53.75	2.12
	5-shot	28.13	2.77	**2.61**	**0.98**	**0.00**	**0.49**	64.50	0.49
	10-shot	28.83	**2.44**	3.09	**0.98**	0.49	0.65	63.19	0.33
	15-shot	28.99	3.58	3.75	1.14	0.33	0.65	61.07	0.49
	fine-tuned	**59.01**	12.05	6.19	**0.98**	0.49	0.98	**20.30**	**0.00**

for verification and trust reasons. This behavior suggests that, without task-specific supervision, models may rely on general medical priors rather than strict document grounding. Fine-tuning substantially mitigated this failure mode by reinforcing extractive behavior and anchoring predictions to the source text. These findings highlight the importance of supervision and grounding constraints for safe deployment of LLM-based clinical QA systems.

Our error taxonomy further helps uncover failure modes that aggregate metrics obscure. By disaggregating errors into categories such as superset, subset, no overlap, and hallucination, we were able to identify concrete patterns of model weaknesses. For example, the prevalence of superset errors suggests that models often err on the side of verbosity, whereas subset errors highlight a tendency to under-extract. Such insights can inform targeted interventions, including modifying prompts, refining training objectives, or designing hybrid pipelines that integrate extractive and generative approaches.

SM acts as a complementary measure of correctness by rewarding models that at least identify the correct sentence containing the gold answer. Its close correspondence to the sum of Exact Match, Superset, Subset, and Overlap in Table IV suggests that many errors arise from boundary misalignment rather than failures of sentence-level localization. This observation highlights practical opportunities for improvement: constraining predicted spans to sentence boundaries, or integrating sentence-level retrieval as a pre-filter, could reduce superset and overlap errors and improve Exact Match accuracy.

The structured error patterns revealed by our taxonomy suggest several concrete algorithmic directions for improving extractive radiology QA. First, persistent boundary errors indicate the potential benefit of sentence-level localization followed by constrained span extraction, where models are first trained or prompted to identify the most relevant sentence before predicting a token-level span. Second, boundary over- and under-extraction could be mitigated through substring-constrained decoding or post-hoc span normalization, restricting outputs to exact substrings of the source report. Third, remaining abstention errors suggest the need for explicit abstention calibration, for example by thresholding model confidence or likelihood ratios when deciding whether to produce an answer versus abstain. While evaluating these strategies is beyond the scope of this benchmark study, our error taxonomy provides a principled framework for guiding and assessing such future algorithmic improvements.

This study is not without limitations. First, we focused on a select set of open LLaMA and Mistral models. We acknowledge that language models pretrained or specialized on medical or radiology-specific corpora may ultimately be better suited for extractive radiology QA, particularly for precise boundary detection and domain-sensitive interpretation. Broader comparisons across diverse model families, including encoder-based clinical transformers and larger closed-source models, would provide a more comprehensive landscape of performance. Second, although RadQA captures clinically motivated questions, it remains limited in size relative to other open-domain QA datasets. Larger and more diverse training corpora may yield further improvements. In addition, RadQA reflects a limited range of radiology report styles and clinical settings. The dataset is derived primarily from a single institution and focuses on specific imaging modalities and report formats, which may limit generalizability to other radiology genres such as impression-only reports, operative notes, or longer and more ambiguous report narratives. Evaluating whether the proposed error taxonomy and observed failure patterns transfer to such heterogeneous report types remains an important direction for future work. Another limitation of this study is the absence of human expert evaluation. While EM, F1, and SM provide objective measures of extraction accuracy and localization, they do not fully capture whether minor boundary deviations alter clinical interpretation or diagnostic meaning. Future work should incorporate radiologist review to assess the clinical acceptability of near-miss predictions, particularly for superset and subset errors. Finally, our evaluation centered on span extraction accuracy; future work should also assess the clinical utility of model outputs, including whether predicted spans support accurate downstream decision-making by physicians. Additionally, while fine-tuning improved performance for the non-reasoning extractive models evaluated here, we note that emerging reasoning-oriented LLMs (often accessible only through few-shot or zero-shot prompting) may exhibit different trends. Evaluating such models was outside

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the scope of this study but represents an important direction for future work.

Taken together, these findings show that fine-tuned LLMs can achieve strong performance on radiology QA while also producing more reliable and clinically interpretable outputs. At the same time, the taxonomy highlights residual failure modes that need targeted strategies, such as reducing boundary errors and further calibrating abstention behavior. Future work should extend this analysis to reasoning-oriented LLMs, which may behave differently under few-shot prompting, and to other clinical QA datasets to validate generalizability.

VI. Conclusion

This work presented a comprehensive benchmark of open large language models for extractive question answering on radiology reports using the RadQA dataset. We evaluated LLaMA 3.1-8B, LLaMA 3.2-3B, and Mistral-Small-22B under zero-shot, few-shot, and fine-tuned settings, and introduced a structured error taxonomy to characterize model behavior. Fine-tuning yielded the largest improvements, with LLaMA 3.1-8B achieving the strongest overall performance at 61% EM, 77% F1, and 85% SM. The taxonomy analysis showed that fine-tuning not only raised accuracy but also reduced severe errors such as hallucination and abstention mistakes, concentrating residual failures in near-miss boundary categories. SM further complemented span-level metrics by highlighting when models located the correct sentence despite imperfect span boundaries.

While fine-tuned extractive LLMs achieve strong results, limitations remain. Residual errors, particularly superset, subset, and overlap cases, suggest the need for better boundary alignment strategies. In addition, reasoning-oriented LLMs, which are often accessible only through few-shot or zero-shot prompting, may exhibit different performance trends. Evaluating such models, along with extending this analysis to other clinical QA datasets and instruction-tuned variants, represents an important direction for future work.

Overall, our results demonstrate that fine-tuned open LLMs can provide accurate and reliable support for radiology report comprehension, moving closer to clinically useful QA systems while also revealing concrete areas for further improvement.

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